VARUN KUSHWAH

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github.com/varunkushwah31 • leetcode.com/u/Varun_Kushwah (500+ Solved) • India

SUMMARY

Computer Science undergraduate specializing in high-concurrency Java backend systems, distributed architectures, and real-time protocols. Proficient in Java 21/25, Spring Boot 3.3, PostgreSQL, Redis, WebRTC, STOMP WebSockets, and Docker. Java & DSA Coordinator at devup mentoring 20+ peers; solved 500+ problems on LeetCode and recognized in Top 105 Nationally in Google Solution Challenge 2025.

EDUCATION

KIET Group of Institutions **2024 – 2028**
B.Tech in Computer Science & Engineering | CGPA: 8.27 / 10.0 *Ghaziabad, India*

New St. Stephens Public School **2021 – 2023**
Senior Secondary (CBSE Class XII: 80%) • Secondary (CBSE Class X: 82%) *Agra, India*

TECHNICAL SKILLS

Languages: Java (17/21/25), SQL, TypeScript, JavaScript, Python

Backend & Frameworks: Spring Boot 3.x, Spring MVC, Spring Security, Spring Data JPA, Hibernate, RESTful APIs, WebSockets, STOMP

Distributed Systems & Networking: WebRTC (RTCDataChannel), Virtual Threads (Java 21/25), Redis 7 (Distributed Locking & Caching), Resilience4j

Databases & Storage: PostgreSQL, MySQL, MongoDB, Redis

DevOps & Tools: Docker, Git, GitHub, Postman, Linux / Bash, Maven, Gradle, CI/CD, Prometheus, Grafana

Core Competencies: Data Structures & Algorithms, Object-Oriented Design (OOP), Microservices Architecture, Concurrency & Multithreading, Clean Architecture

EXPERIENCE & LEADERSHIP

devup — College Tech Club **Aug 2024 – Present**
Java & DSA Coordinator • Technical Mentor *KIET Campus*

- **Java Backend Curriculum:** Spearheaded structured hands-on Java & Spring Boot backend modules for **20+ active student members**, preparing juniors for software engineering internships and technical assessments.
- **DSA Mentorship:** Conducted weekly algorithmic problem-solving sessions on trees, graphs, and dynamic programming, improving juniors' average coding speed and benchmark accuracy by **30%**.
- **Engineering Standards:** Orchestrated code reviews and Git workflows, enforcing clean architecture patterns, OOP principles, and test-driven practices across collaborative cohort projects.

PROJECTS

MentorSync | LeetCode & Codeforces LMS Live Demo | GitHub
Java 21, Spring Boot 3.3, React 19, TypeScript, Redis 7, MongoDB, STOMP WebSockets, Resilience4j, Docker

- **Dual-Platform Telemetry:** Architected a scalable two-sided LMS for bootcamps to automate LeetCode and Codeforces progress tracking, assignment distribution, and real-time leaderboards.
- **Automated Verification Engine:** Built an asynchronous verification engine consuming LeetCode GraphQL and Codeforces REST APIs with problem slug parsing, eliminating manual grading across **100+ active submissions**.
- **Cache Warming & Resilience:** Engineered nightly Redis cache-warming worker (03:00 UTC) with atomic distributed locks (`setIfAbsent`) ensuring **0ms peak latency**, STOMP WebSockets, and Resilience4j circuit breakers.

W2W Share | Offline Encrypted P2P File Transfer Live Demo | GitHub
Java 25, Spring Boot 4.1, WebRTC, Bouncy Castle, Virtual Threads, ZXing, Docker, Prometheus

- **Zero-Relay P2P Protocol:** Engineered an offline peer-to-peer file transfer platform on WebRTC RTCDataChannels over UDP that bypasses central servers with zero internet requirement.
- **High-Concurrency Signaling Broker:** Built a Java 25 Virtual Threads and WebSocket signaling broker supporting **10,000 pooled connections**, with ZXing QR pairing and UDP subnet auto-discovery.
- **Encrypted Chunk Streaming:** Designed a 64KB chunk streaming protocol with Bouncy Castle AES-256-GCM encryption and backpressure control, handling up to **50GB transfers with <50MB of RAM**.

ACHIEVEMENTS

- **500+ Problems Solved on LeetCode:** Demonstrated strong algorithmic mastery across arrays, strings, trees, dynamic programming, and graph algorithms (leetcode.com/u/Varun_Kushwah).
- **Top 105 Nationally — Google Solution Challenge 2025:** Recognized out of thousands of submissions across India for engineering real-world social impact software.
- **2nd Place — Innotech (Department Level):** Awarded for technical innovation in high-throughput backend architecture.